

Statement of Purpose

In Bangladesh, where I grew up, a misdiagnosis does not come with an easy appeal process. Hospitals in smaller cities are understaffed and overloaded, and patients often have one chance to get a reading right. When I first started thinking about machine learning, this is where my mind went: not to optimization benchmarks, but to whether a model could actually be trusted where the stakes are real. That question became concrete when I built my own insurance risk prediction system and realized that even a model with strong holdout accuracy could not explain a single one of its predictions. The gap between a working model and a trustworthy one turned out to be enormous.

Reading Asst. Prof. Polat Goktas's paper "Shaping the Future of Healthcare: Ethical Clinical Challenges and Pathways to Trustworthy AI" (JCM, 2025) answered a question I had long held: if AI performs so well in clinical tasks, why has it not been broadly adopted in healthcare? Goktas makes the risk clear. Clinical AI models often function as opaque black boxes, and CNNs in medical imaging illustrate this directly: capable of detecting tumors in MRI scans with high accuracy, yet frequently unable to explain their predictions in ways clinicians can interrogate or trust. That section of the paper stayed with me most, and it has since deepened my motivation to pursue research in AI for healthcare.

I am Shafikul Islam Marwan, a third-year CSE student at Jashore University of Science and Technology with a CGPA of 3.78 out of 4.00. I have solved over 2,100 data structure and algorithm problems on various competitive programming platforms, participated in 100+ online programming contests, achieved the rank of Specialist on Codeforces, won the JUST intra-University Programming Contest in 2025, and completed a six-month machine learning course with Phitron at 90.63% grade points. These have built both a problem-solving discipline and a practical grounding in the full ML pipeline.

I have deployed two complete systems: an insurance premium predictor using Python, scikit-learn, FastAPI, and Streamlit, and a behavioral personality classifier that reached 92.93% test accuracy and runs live on Hugging Face Spaces. Both pushed me toward questions that accuracy metrics do not answer. A model that labels someone high-risk without any explanation is not just incomplete. It is a liability. That conviction connects directly to the research I want to pursue here.

My top two preferences are Prof. Goktas's projects: AI for Open Innovation in Life Sciences and Building Sustainable Smart Cities with AI and IoT. Both center on building ML systems that are interpretable as well as accurate, using open data and reproducible pipelines. For the life sciences project, I want to work specifically with explainability tools such as SHAP and LIME to examine whether performance benchmarks alone measure a model's fitness for clinical use. My third and fourth choices are the genome-wide UV damage analysis and phylogenetic atlas projects under Assoc. Prof. Ogun Adebali. Both are fully computational, and the skills they require, handling large-scale sequencing datasets and building reproducible analysis pipelines, map closely to what I already practice in Python.

I have no published research, and I am not pretending otherwise. What I do have is a genuine understanding of why the problems PURE addresses matter, the technical foundation to contribute from day one, and a clear direction: a career in applied AI built not just on what models can do, but on when they deserve to be trusted. PURE is where that foundation gets built properly.